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Morphology-Aware Tokenization for Malayalam: Design and Evaluation

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List of Abbreviation

NLP : Natural Language Processing

G2P : Grapheme-to-Phoneme

PII : Personally Identifiable Information

JSONL : Javascript Object Notation Lines

SOV : Subject–Object–Verb

OSV : Object–Subject–Verb

BRAT : brat rapid annotation tool

LOLI : lexically-only items

LIBSVM library for support vector machines

Summary

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Abstract

This is Abstract

Literature Review

The contemporary evolution of natural language processing (NLP) for Dravidian languages is increasingly defined by the profound computational tension between universal subword tokenization algorithms and the highly specific structural requirements of agglutinative morphologies. As the architectural foundation of artificial intelligence continues to shift toward massive, multilingual Large Language Models (LLMs) and Transformer-based frameworks, the methodologies used to represent raw text to these neural networks have become a critical locus of academic inquiry. Tokenization, the fundamental preprocessing step that defines the discrete units over which neural models operate, dictates the vocabulary, context window utilization, and ultimate semantic comprehension of the system. For high-resource, fusional languages such as English, standard subword algorithms have substantially reduced traditional OOV problems in English and other high-resource settings.[1] However, applying these identical algorithms to structurally dense, low-resource languages introduces severe representational biases.

Malayalam, a prominent Dravidian language spoken by approximately 38 million individuals primarily in the Indian state of Kerala and its global diaspora, represents a critical and highly challenging case study within this computational domain. Typologically, Malayalam features an intricate morphological framework defined by profound agglutination, complex sandhi (morphophonemic) phenomena, and extensive inflectional mechanisms. Unlike analytic or fusional languages, where grammatical relationships are predominantly signaled by strict word order or a limited set of free-morpheme prepositions, Malayalam encodes vast amounts of syntactic and semantic information within a single, continuous orthographic word. This dense structural capacity inherently generates an exceptionally high type-to-token ratio, yielding word forms that are characteristically long, rare, and highly inflected.[2][3]

This agglutinative nature induces a fundamental tension in text representation. Under-segmentation—treating these long agglutinative forms as single, atomic tokens—leads to unwieldy vocabulary sizes and extreme data sparsity, which fundamentally cripples the learning capacity of neural models. Conversely, the prevailing industry standard of data-driven subword tokenization, primarily implemented through algorithms such as Byte Pair Encoding (BPE), WordPiece, or Unigram Language Modeling, frequently results in pathological over-segmentation. Because these algorithms are fundamentally agnostic to linguistic boundaries and operate almost exclusively on statistical character co-occurrence frequencies, they routinely fragment morphologically coherent units into semantically opaque, meaningless character n-grams. This fragmentation can obscure linguistically meaningful morpheme boundaries, reducing representational efficiency and increasing sequence length.[1], [4]

The resulting sequence inflation, commonly theorized in recent literature as the "token tax," exacts a severe toll on model performance and computational efficiency. The self-attention mechanism central to Transformer architectures operates with a computational complexity of $O(n^2)$ relative to the sequence length; thus, unnecessarily fragmented tokens exponentially increase memory allocation and processing time while simultaneously diluting the semantic density of the context window[5], [6]. Consequently, recent work in Malayalam language modeling shows growing interest in hybrid, morphology-aware approaches that aim to balance computational efficiency with linguistic fidelity.[4]

This literature review systematically examines the foundational linguistic complexities of Malayalam, the theoretical frameworks and inherent limitations of prevailing subword algorithms, and the recent developments in morphology-aware tokenization in recent years.

0.1 The Morphological Landscape of Malayalam

The typological classification of Malayalam as a highly agglutinative language constitutes the primary theoretical barrier to the implementation of standard subword tokenization methodologies. In agglutinative structural frameworks, surface word forms are constructed by linearly concatenating a root morpheme with multiple affixes, each of which typically carries a single, distinct piece of grammatical or semantic information. A single Malayalam word can encode a complex, hierarchical array of linguistic features, including case, number, person, tense, aspect, mood, and various postpositions. Consequently, a single orthographic unit in Malayalam frequently corresponds to an entire multi-word phrase or dependent clause in an analytic language such as English.

This morphological density dictates that noun inflection in Malayalam adheres to a rigid, highly productive sequence of affixation. The nominal root forms the structural core, which remains relatively constant in semantic meaning. This root is sequentially followed by gender markers, plural markers, case suffixes, and finally, terminal postpositions or clitics. The Malayalam case system is exceptionally dense and functionally vital, typically marking seven to nine distinct cases that convey complex grammatical relations.

Case Category	**Typical Suffix**	**Semantic Function**	**Example Inflection**
Nominative	(Null)	Subject	maraM (tree)
Accusative	-e, -ye	Direct Object	maratthe (the tree)
Dative	-inu, -nu, -u	Indirect Object	maratthin (to the tree)
Instrumental	-al	Means/Instrument	maratthal (by the tree)
Sociative	-otu	Association	maratthotu (with the tree)
Locative	-il, -kal	Location	maratthil (in the tree)
Genitive	-ute, -nte	Possession	maratthinte (of the tree)

Table 1: Malayalam Case Markers and Their Functions in English Transiliteration

The precision of computational tokenization at these specific morphological boundaries is often important for Natural Language Understanding (NLU). When a statistical tokenizer arbitrarily fractures a case marker, it effectively deletes the vital grammatical signal that dictates whether a specific noun operates as the agent, the recipient, or the location of an action within the sentence's dependency parsing tree. [7], [8], [9]

Verbal morphology in Malayalam exhibits an even greater degree of agglutinative complexity. Verb forms systematically incorporate markers for tense (past, present, future), aspect (perfective, progressive), and mood (indicative, imperative, conditional). These suffixes are stacked in a highly specific structural hierarchy where tense typically acts as the final mandatory marker before the attachment of any closing clitics. For instance, the verb root varuka (to come) can generate highly inflected surface forms such as vannittilla

(has not come).This surface form represents a complex fusion of the past stem, a perfective aspect marker, and a terminal negative clitic. A purely statistical, morphology-blind tokenizer is mathematically predisposed to fracture this form into arbitrary subunits (e.g., [van, ni, tti, lla]), thereby failing entirely to preserve the isolated negative particle illa. A linguistically aligned, morphology-aware tokenization pipeline would ideally isolate the functional morphemes (e.g., [vanni, ttu, illa]), preserving the syntactic clarity required for downstream neural classification. Furthermore, the syntactic flexibility of Malayalam, which permits both Subject-Object-Verb (SOV) and Object-Subject-Verb (OSV) sequence orders, places the absolute burden of semantic meaning onto these morphological markers, making their preservation a relevant design consideration.

0.2 Sandhi Rules and Morphophonemic Alternations

A critical theoretical challenge in processing and tokenizing Malayalam text is the pervasive presence of productive sandhi rules. In computational linguistics, sandhi refers to the morphophonemic alternations or euphonic phonological changes that naturally occur at the juncture of two morphemes during agglutination.These rules dictate the necessary insertion, deletion, or transformation of transitional characters to ensure the phonetic ease of pronunciation. For example, the addition of the locative case suffix -il to the nominal root kāṭu(forest) results in the surface form kāṭṭil, demonstrating the productive phonological doubling of the "t" character.

From a computational modeling perspective, sandhi poses a parameter-sharing problem. If a tokenizer does not explicitly account for these euphonic alternations, the underlying neural model is forced to treat the base root kāṇu and the sandhi-altered stem kāṭṭ- as entirely disparate lexical entities. This representational failure prevents the model from sharing the learned embedding parameters associated with the semantic concept of "forest" across its various inflected contextual forms. The inability to systematically align sandhi-altered stems is identified in current literature as a primary mechanism by which standard Byte Pair Encoding underperforms in agglutinative contexts. Consequently, theoretically sound tokenization pipelines must be "sandhi-aware," either by reversing these phonological changes during a deterministic pre-processing step or by ensuring that the subword vocabulary robustly includes the varied forms of roots and suffixes produced by these euphonic rules.

0.3 Orthographic Evolution and Digital Representation Challenges

The digital processing of Malayalam is inextricably linked to, and further complicated by, its orthographic history. The Malayalam script, classified as an alphasyllabary or abugida derived from the ancient Brahmi script via Grantha, historically utilized a highly complex typographical system of consonant-consonant and consonant-vowel ligatures. In 1971, the Government of Kerala implemented a comprehensive orthography reform strictly aimed at simplifying the script for modern typographic and printing technologies.[10] This reform aggressively reduced the overall number of complex ligatures and standardized certain vowel and consonant combinations.

However, this historical shift has resulted in a lasting, highly problematic dichotomy in contemporary digital text representation. Modern computational corpora frequently

contain a heterogeneous mixture of both traditional and reformed orthographic practices. This generates multiple distinct graphical realizations for the exact same underlying linguistic sequence. This phenomenon is further exacerbated by varied and inconsistent Unicode usage across different platforms and eras; older digital documentation and modern user-generated web content may encode identical graphemes utilizing completely different byte sequences. Specifically, the erratic encoding and visual representation of virama and chillu characters (special independent consonants distinct to Malayalam) frequently confounds basic normalization heuristics and subword tokenizers operating at the character level. Without rigorous Character Normalization applied as a core part of the experimental design, visually identical but byte-divergent texts will generate a fragmented, highly inflated tokenizer vocabulary where the exact same phonological sound is represented by multiple disparate subword tokens.

Furthermore, spacing practices in Malayalam digital text often deviate significantly from strict lexical boundaries. While formal modern writing theoretically separates words with spaces, the absolute number of characters between spaces remains extremely high in agglutinative constructions. In informal digital registers—such as social media discourse, conversational blogs, and YouTube comments—whitespace frequently corresponds to phonological breath pauses rather than strict morphological or lexical divisions. Writers exhibit high variance in their spacing conventions around clitics, particles, and postpositions, rendering naive whitespace tokenization highly unreliable. Any robust attempt at tokenization should consider that whitespace is not a definitive indicator of a token boundary, necessitating deeper morphological analysis to prevent the inflation of vocabulary through spurious token types.

0.4 Intra-Word Code-Switching and Morphological Adaptation

A defining, modern characteristic of Malayalam digital dialogue and informal web text is the high prevalence of code-switching, specifically English-Malayalam intra-word switching. This profound linguistic hybridity occurs when English lexical roots are directly subsumed into Malayalam's agglutinative morphological framework, necessitating immediate phonological and morphological adaptation to ensure grammatical consistency within the sentence.

Documented examples in conversational corpora reveal patterns of asymmetric morphological incorporation where English roots serve as the invariant base for Malayalam suffixation. Examples include hybrid forms such as "director-inte" (combining the English root "director" with the Malayalam possessive genitive suffix "-inte") and "point-ukal" (combining the English root "point" with the Malayalam plural suffix "-ukal", which uniquely necessitates the insertion of an epenthetic vowel 'u' to bridge the phonological gap between the English terminal consonant and the Malayalam marker). This asymmetry reflects the sociolinguistic prestige of English as a primary source of technical and modern vocabulary in Kerala.

For data-driven tokenization algorithms, these hybrid linguistic units are highly problematic. Because standard subword tokenizers are predominantly trained on monolingual, formal data distributions, they routinely classify these cross-linguistic blends as "unknown" statistical sequences, fragmenting them into arbitrary, meaningless pieces rather than isolating the semantic English root from the functional Malayalam suffix. Developing robust NLP systems capable of operating on realistic, code-mixed text may ben-

efit morphology-aware hybrid tokenizers capable of incorporating these cross-linguistic "sandhi" rules into the subword segmentation process.

0.5 Methods, Approaches, and Technologies

0.5.1 Theoretical Frameworks of Subword Tokenization Algorithms

The computational transition from word-level tokenization to subword tokenization defines the current era of neural sequence modeling, driven by the necessity to balance representational coverage for infinite word forms against computational vocabulary efficiency. However, the foundational mathematical mechanisms of these algorithms heavily dictate their disparate efficacy in agglutinative environments.

Byte Pair Encoding (BPE) and WordPiece:[11] Originally developed as an algorithm for data compression, Byte Pair Encoding (BPE) was adapted for NLP to enable open-vocabulary modeling and effectively eliminate the out-of-vocabulary (OOV) problem. BPE operates through a greedy, deterministic, and iterative merging heuristic. It initializes with a basic character-level vocabulary and progressively merges the most statistically frequent adjacent symbol pairs in the training corpus until a predefined, fixed vocabulary size (e.g., 32,000 tokens) is achieved. While highly efficient for high-resource, fusional languages with concatenative morphology like English, BPE merge decisions are driven entirely by raw statistical frequency rather than underlying morpheme structure. In the context of Malayalam, this frequently results in splits that cut indiscriminately across critical semantic units, especially within long nominal and verbal suffix chains.WordPiece utilizes a fundamentally similar iterative merging logic but optimizes for a likelihood-based objective. It calculates a mathematical score for each character pair based on its joint frequency relative to the product of its individual frequencies, aiming to identify subwords that strictly maximize the probability of the training data. During inference, WordPiece employs a longest-match-first strategy. Like BPE, it prioritizes statistically frequent character clusters that rarely align with Malayalam's functional morphemes, rendering both algorithms highly prone to pathological fragmentation.

Unigram Language Modeling and SentencePiece:[12] The Unigram language model algorithm reframes tokenization from a deterministic merge process to a probabilistic generative process. Initializing with a massive, over-provisioned vocabulary, Unigram iteratively prunes subwords that demonstrate low likelihood using the Expectation-Maximization (EM) algorithm. Given an input string x Unigram utilizes Viterbi dynamic programming to search for the optimal segmentation s^* that explicitly maximizes the joint probability:

$$P(s) = \prod_{j=1}^{n} P(v_{i_j}) \tag{1}$$

where v represents tokens from the vocabulary. This probabilistic approach uniquely allows for multiple valid segmentations of the same word, making it generally more flexible and often slightly more linguistically aligned than rigid, deterministic merge-based methods like BPE.SentencePiece, a prominent and widely adopted implementation architecture, can utilize either BPE or Unigram objectives and treats whitespace simply as a

regular character, enabling theoretically language-independent processing. However, despite the mathematical sophistication of the Unigram model, empirical research confirms that it still struggles profoundly with the extreme type-to-token ratios characteristic of Malayalam. Without explicit morphological guidance, it frequently fails to achieve optimal accuracy, still producing unnatural splits in heavily agglutinated sequences.

0.5.2 The "Token Tax" and Representational Bias

The fundamental misalignment between statistical likelihood and linguistic integrity in agglutinative languages imposes severe computational and semantic penalties, conceptualized in current literature as the "token tax" or "token inflation".

Feature	**Malayalam (Agglutinative Typology)**	**English (Fusional/Analytic Typology)**
Primary Grammatical Strategy	Agglutination with suffix stacking	Syntax-driven structure using word order and prepositions
Average Word Length	Long, typically multi-morphemic and agglutinated	Short, mostly atomic morphemes
Morphological Density	High (multiple morphemes per word)	Low (few morphemes per word)
Stem Stability	Variable due to complex sandhi changes	Relatively stable
Tokenization Challenge	Excessive fragmentation and token inflation	Minimal fragmentation

Table 2: Comparative theoretical impact of morphological typology on tokenization behavior

High fragmentation—quantified through specific intrinsic metrics—forces a neural model to attempt to reconstruct complex semantic meaning from opaque, fragmented sub-units rather than processing coherent, recognizable morphemes. More critically, this fragmentation exponentially inflates the input sequence length. Because the core self-attention mechanism in modern Transformer architectures operates with a computational complexity of $O(n^2)$ relative to the number of tokens, evaluating a heavily fragmented Malayalam sentence requires disproportionately higher memory allocation, computational compute, and processing time compared to evaluating the exact same semantic content in English. In morphologically rich languages such as Malayalam, standard BPE tokenization often produces elevated subword fertility, as complex inflected word forms are fragmented into multiple smaller units rather than preserved as linguistically coherent segments [13], [14]. increases in token fertility are associated with lower downstream model accuracy: in one recent study, each additional token per word corresponded to an average accuracy decline of 8 to 18 percentage points, depending on the model and task [15]. This establishes token fragmentation as a primary, systemic barrier to both high performance and economic feasibility in Dravidian NLP, effectively "wasting" the model's limited context window.

To objectively quantify these cross-lingual representational biases, current methodologies rely on specialized parity and efficiency metrics.

Tokenization Parity (TP) evaluates the ratio of tokenized lengths for semantically parallel content across different languages. Recent analyses of tokenization parity suggest that multilingual models frequently encode Dravidian and other morphologically rich Indic languages less efficiently than English. In practice, this is reflected in TP values greater than 1, which correspond to longer tokenized sequences and therefore higher computational cost and reduced effective attention capacity for the same semantic content [14], [16].

Information Parity (IP) assesses the relative compression efficiency of an LLM for a given language relative to a reference language. Recent scholarly research demonstrates that IP serves as a significantly stronger and more robust predictor of downstream performance in semantic reasoning tasks—such as topic classification—than simple token fertility counts, as it isolates the semantic density of the representation from mere token volume.

0.5.3 Intrinsic Evaluation Metrics for Tokenization Quality

To objectively benchmark tokenizer efficiency, compression, and linguistic alignment independent of any specific downstream task, researchers rely on a structured suite of intrinsic evaluation metrics. These metrics provide a fundamental, language-agnostic methodology to diagnose representational quality before proceeding to expensive neural model training:

1. **Fertility:** The mathematical average of subword tokens produced per word in a reference corpus. It serves as the primary proxy for sequence length inflation and computational cost.

2. **Word Fragmentation Rate (WFR):**The direct percentage of total words that are split into more than one token. A high WFR signals the tokenizer's failure to capture whole words or large, meaningful grammatical stems, forcing reliance on sub-lexical fragments.

3. **Characters Per Token (CPT):** A metric indicating token granularity and compression; higher CPT values strongly suggest that the tokens represent larger, more meaningful linguistic units (such as complete stems or terminal affixes) rather than arbitrary single characters.

4. **Single Token Retention Rate (STRR):** The specific proportion of high-frequency reference words (often derived from a gold-standard linguistic dictionary) that are successfully preserved as single, atomic tokens by the algorithm. A low STRR for Malayalam compared to English implies a systemic, structural bias in the tokenizer's vocabulary allocation budget.

5. **Morph-Boundary Violation Rate:** Regarded as the most direct measure of linguistic fidelity. It quantitatively compares the algorithmic token boundaries predicted by the tokenizer against a gold-standard set of true morphological splits defined by linguists. This is strictly evaluated using standard classification metrics: Precision, Recall, and the harmonic mean (F1-score) to evaluate absolute boundary alignment.

0.5.4 Lightweight Transformer Architectures for Malayalam Deployment

The demand for efficient, highly scalable AI in low-resource environments has precipitated the rapid development of lightweight Transformer models. Designed to provide robust performance with a mere fraction of the parameter count and computational overhead of massive LLMs, these models are the ideal deployment targets for Malayalam NLP. However, due to their reduced capacity (fewer layers, smaller embedding dimensions), lightweight Transformers are exceptionally sensitive to the quality of their input representation. If tokenization is highly fragmented, the limited representational capacity of the small model is entirely wasted on the syntactic reassembly of fractured words, rather than deducing higher-order contextual and semantic relationships.

Prominent lightweight architectures utilized for benchmarking Malayalam NLP include:

IndicBERT v2: IndicBERT v2 is highly relevant to this research because it offers a strong multilingual Transformer backbone for Indic language understanding. Pretrained on IndicCorp v2, which spans 24 Indic languages including Malayalam, it can be adapted to downstream tasks through standard fine-tuning or lighter parameter-efficient methods such as LoRA when computational efficiency is important [17], [18].

ALBERT: Utilizes cross-layer parameter sharing techniques to achieve an extremely low memory footprint, making it highly ideal for resource-constrained production environments.

DistilBERT: Employs advanced knowledge distillation techniques, achieving an architectural footprint that is 40% smaller and 60% faster than the standard BERT-base model, while successfully retaining 97% of its accuracy. [19]

MiniLM: Deeply optimized via distillation and specific pruning methodologies to achieve extremely high throughput and ultra-low latency inference, specifically targeted for edge devices.

The interaction between these lightweight models and tokenization is highly synergistic. By effectively reducing the absolute number of tokens per word through morphology-aware segmentation, the effective context window of the lightweight model is dramatically expanded, parameter waste is minimized, and inference latency is significantly reduced.

0.6 Advancements in Morphology-Aware Tokenization

The chronological epoch in the past years has witnessed a marked, transformative scholarly progression toward explicitly linguistically informed preprocessing paradigms for Dravidian languages. A highly notable advancement within this timeframe is the development of **MorphBPE**[20], a morphology-aware extension of the foundational BPE algorithm, introduced in late 2024 and subsequently refined in early 2025 by Sheth et al.Importantly, MorphBPE is a broadly applicable tokenizer design rather than a Malayalam-specific controlled study. . MorphBPE's technical architecture fundamentally modifies the core BPE merge criterion by directly incorporating explicit morpheme boundary annotations into the statistical learning phase. While vanilla BPE merges character pairs based exclusively on raw frequency

$f(c_i c_j)$, MorphBPE calculates a mathematically constrained merge score S_λ that actively rewards within-morpheme merges while strictly penalizing any proposed merge

that crosses an annotated morphological boundary:

$$S_\lambda(c_i, c_j) = f(c_i c_j) + \lambda\, m(c_i, c_j) - \mu\, b(c_i, c_j) \tag{2}$$

Empirical evaluations conducted on highly agglutinative languages demonstrate that MorphBPE preserves meaningful morphological units, improves alignment with true morpheme boundaries, reduces cross-entropy loss, and accelerates training convergence relative to standard BPE. In the reported experiments, it reached a reference validation loss using up to 29% fewer training tokens than unconstrained statistical baselines, with particularly strong gains for morphologically rich languages [20].

Concurrently, the **IndicSuperTokenizer** established a new benchmark for subword modeling in the Indic multilingual setting by combining subword and multi-word tokenization with language-specific pre-tokenization rules, yielding more linguistically aligned token boundaries and improving average fertility by 39.5% over LLaMA4-based tokenizers [13]. Related work on Telugu further showed that hybrid tokenization strategies incorporating morphological pre-segmentation can improve performance within BPE-based frameworks, although tokenizer choice itself remains a major factor in downstream results [21].

0.7 Algorithmic Dominance, Alignment, and Predictability

Despite these advancements, the computational linguistics community continues to rigorously debate the exact mechanisms and algorithms driving optimal downstream performance in morphologically rich languages. A pivotal 2024 study by Klein and Tsarfaty ("Rethinking tokenization for rich morphology") argued for the general theoretical dominance of the probabilistic Unigram algorithm over deterministic BPE in successfully capturing true morphological alignment without explicit rule-based interventions.

However, subsequent large-scale research published recently, encompassing an expansive evaluation across 70 distinct languages, provided a significantly more nuanced perspective. The study concluded that while morphological alignment—defined as the exact mathematical degree to which predicted subword tokens correspond to true, linguistically valid morphemes—was indeed a highly significant predictor for performance in structure-sensitive tasks (such as Dependency Parsing and POS tagging), it did not universally dictate success. The overarching algorithm chosen (BPE versus Unigram) often remained a heavier, independent influence on aggregate model performance than the degree of morphological alignment alone, suggesting complex interactions between token distribution and neural attention mechanisms.

Furthermore, critical research investigating tokenization fairness and cross-lingual bias (Ahia et al., 2025) highlights that representational metrics strictly dictate downstream task success. Their findings reveal that Tokenization Parity (TP) correlates strongly with span-based and syntactic tasks, whereas Information Parity (IP) serves as the primary, most robust predictor for tasks demanding deep semantic understanding and reasoning, such as topic classification.

0.8 Datasets, Benchmarks, and Downstream Task Performance

The acquisition, curation, and standardization of high-quality, annotated datasets remain a persistent bottleneck for Malayalam NLP, prompting significant recent collaborative

curation efforts.

- **SIB-200 (2024):** Developed by Adelani et al. and based on the Flores-200 multi-way parallel corpus, SIB-200 provides an essential, large-scale evaluation benchmark specifically for topic classification across 200+ languages. It includes approximately 1,000 highly curated, category-labeled samples for Malayalam, serving as a primary standard for cross-lingual NLU evaluation. [22]
- **Dravidian LangTech Shared Tasks:** Released via continuous, dedicated workshops at major conferences such as NAACL and EACL, these datasets have established the definitive gold standard for specific downstream tasks in Malayalam, yielding critical, balanced data for fake news detection (4,000 samples)[23] and complex offensive language classification.[24]
- **Topic and Language ID Corpora:** Recent open-access resources have expanded the landscape of Malayalam text classification and code-mixed language identification. For topic classification, Malayalam news corpora are now available through broader Indic news benchmarks, although dataset size and category composition vary across releases [25]. For language identification, the WoLLaI Malayalam-English dataset provides a dedicated benchmark for word-level labeling in code-mixed text, comprising 12,502 annotated sentences with tags for Malayalam, English, mixed forms, and other tokens [26].

Contemporary models evaluated on these robust datasets—specifically pre-trained, Indic-focused multilingual transformers like MuRIL, XLM-ROBERTa, and IndicBERT v2—have demonstrated exceptional capabilities. Recent work on Malayalam fake-news detection using social-media and YouTube-comment datasets has shown that fine-tuned transformer models can achieve strong binary-classification performance, with reported macro F1 scores in the range of approximately 0.86–0.88 [27], [28]. Performance on multiclass formulations, however, remains notably lower and more variable across systems [29]. Complementary explainability studies using tools such as LIME, Anchor, and Occlusion suggest that transformer-based Malayalam models rely on linguistically meaningful cues, although broader claims about superior morpho-syntactic modeling over general multilingual baselines should be made cautiously [30].

Future algorithmic outlooks projected with past five year timeframe suggest a rapid transition away from rigid static rules toward Neural Morphological Segmenters—small, highly optimized neural models trained specifically to identify morpheme boundaries and handle complex sandhi fluidly as a pre-tokenization step—and Syllable-Byte Pair Encoding (S-BPE), which biologically grounds subword modeling by forcing segmentation exclusively at valid phonetic pronunciation boundaries.

0.9 Research Gaps and Motivation

Despite the rapid and impressive proliferation of multilingual Large Language Models and highly advanced subword algorithms over the past five years, a critical representational gap remains in the processing of agglutinative languages. The contemporary literature comprehensively demonstrates that purely data-driven algorithms like Byte Pair Encoding and WordPiece fundamentally lack linguistic awareness, resulting in the aggressive, statistically driven fragmentation of Malayalam's dense suffix chains, complex compound verbs, and sandhi-altered roots. As evidenced conclusively by the human-centered evalua-

tions conducted in previous studies, this pathological over-segmentation actively destroys the semantic coherence of inflected forms, divorcing functional grammatical markers from their lexical stems. Conversely, while traditional rule-based tokenizers excel at preserving structural integrity and achieving high human interpretability scores, they are rigidly constrained by closed, finite vocabularies. As a result, purely rule-based systems fail to efficiently scale to massive neural modeling tasks and inevitably falter when confronted with the ubiquitous out-of-vocabulary terms continuously generated by modern digital code-switching and informal web discourse.

Recent theoretical innovations within the 2024–2026 timeframe, such as MorphBPE and the IndicSuper Tokenizer, clearly indicate that the integration of explicit morphological constraints into statistical learning provides a highly effective middle ground. These hybrid models successfully minimize the "token tax" and accelerate training convergence while concurrently maintaining the high vocabulary coverage essential for deep learning. However, despite these architectural advancements, there remains a distinct scarcity of rigorously controlled, reproducible empirical studies that explicitly isolate the specific downstream impact of a lightweight, rule-based "morphological shield" applied as a pre-tokenization constraint layer on standard BPE tokenization within the Malayalam context. Specifically, it remains undetermined to what exact statistical extent pre-tokenization boundary preservation targeting precisely defined, high-frequency phenomena like nominal case suffix stacking and verbal inflectional endings—directly influences downstream Natural Language Understanding tasks.

Topic classification serves as an optimal, highly sensitive stress test for evaluating this tokenization paradigm. Topic classification strictly requires a neural model to neutralize surface-level inflectional variance to accurately identify underlying, shared semantic lemmas across documents. If a tokenizer inconsistently splits a core morphological root across various inflected contexts due to a lack of sandhi-awareness or boundary protection, the neural model wastes its limited parameter capacity attempting to reconstruct that shared semantic identity from disparate, fragmented character strings.

Therefore, a significant research gap exists in systematically quantifying the correlative relationship between intrinsic tokenization quality metrics (specifically the Word Fragmentation Rate, average token fertility, and morph-boundary violation rate) and downstream neural topic classification performance under strictly controlled, fixed-vocabulary conditions. This limitation in existing Malayalam NLP research motivates the methodological approach adopted in this study, which formalizes a constrained morphological preservation layer to critically evaluate the algorithmic trade-offs between linguistic boundary preservation and computational efficiency, as discussed in the following chapter.

Algorithm	Merging Principle	Linguistic Awareness	Best Use Case
Standard BPE	Frequency-based greedy merge	None	High-resource analytic languages
Unigram	Probabilistic pruning (EM)	Latent/Probabilistic	Multilingual general-purpose models
MorphBPE	Frequency with boundary penalty	High (explicit morpheme constraints)	Agglutinative and templatic languages
IndicSuperTokenizer	Multi-grain tokenization with pre-tokenization	High (language-specific rules)	Low-resource Indic and Dravidian languages

Table 3: Comparison of tokenization algorithms

0.9.1 The Morphology-Aware Hybrid Framework

To mitigate the computational token tax and semantic distortion inherent in standard BPE, emerging theoretical methodologies propose the direct integration of linguistic structure into the subword learning process through hybrid tokenization. The "Morphological Shield" is a formalized, rule-based pre-tokenization constraint layer meticulously designed to enforce linguistic alignment without requiring the computationally prohibitive step of a full, deep morphological analysis for every word.

The shield mechanism acts as a protective boundary layer targeting the most highly frequent and regular agglutinative structures, operating through three primary, sequential heuristics:

1. **Terminal Suffix Stripping:** Utilizing formalized rules to identify high-frequency, word-final clitics and nominal case markers, subsequently inserting a virtual computational boundary to separate them from the core stem.
2. **Boundary Preservation:** Utilizing a strictly constrained set of regular expressions (regex)—often incorporating complex lookbehind and lookahead assertions—to definitively prohibit the subsequent BPE algorithm from merging characters across these established virtual boundaries. This intervention specifically ensures that highly productive, functionally critical suffixes (e.g., the genitive -uTe or the plural -kal) are treated as inviolable, atomic units by the statistical learner.
3. **Sandhi Normalization:** Applying simple, reversible, rule-based phonological transformations to restore altered roots to a standardized, common base form prior to vocabulary training. This step dramatically maximizes parameter sharing and consistency in the learned embeddings by ensuring the algorithm recognizes disparate surface forms as originating from the identical semantic root.

In practice, the raw Malayalam text is first passed through this linguistic shield layer. The resulting "pre-tokenized" corpus which now contains explicit linguistic markers at

Metric	Definition	Malayalam Context
Word Fragmentation Rate (WFR)	Percentage of words split into more than one token	High in BPE due to complex suffixation
Average Token Fertility	Number of tokens generated per word form	Crucial for sequence length optimization
Tokenization Parity (TP)	$t(s_{\text{Malayalam}})$	Measures balance across tokenization strategies
Information Parity (IP)	Relative compression efficiency	Predicts semantic task performance

Table 4: Evaluation metrics for tokenization in the Malayalam context

morphologically significant boundaries—is then utilized to train the standard BPE model. Because the data - driven BPE algorithm is technically forced to respect these pre - established boundaries, the resulting subword vocabulary naturally and mathematically prioritizes semantic morphemes over statistically frequent but linguistically meaningless character strings.

1. Research Design

This study adopts a controlled comparative experimental design to evaluate the impact of morphology-aware tokenization on Malayalam natural language processing. The objective is to examine whether incorporating linguistic structure into the tokenization process improves both representation quality and downstream task performance when compared to a standard data-driven approach.

The experiment compares two tokenization strategies under identical conditions: (1) a baseline Byte Pair Encoding (BPE) tokenizer, and (2) a morphology-aware hybrid BPE tokenizer enhanced with a rule-based constraint layer ("Morphological Shield").

The central hypothesis of this study is twofold. First, it is expected that the integration of morphological constraints will reduce subword fragmentation and improve alignment with true linguistic boundaries. Second, it is hypothesized that these improvements in representation will translate into better performance on a downstream classification task.

To ensure a valid comparison, all non-target variables are strictly controlled. Both tokenizers are trained on the same standardized Malayalam corpus, with an identical vocabulary size of 32,000 tokens. The downstream evaluation is conducted using the same neural architecture (IndicBERT v2), identical training configurations, and fixed dataset splits. This "minimal pair" setup ensures that any observed differences in performance can be attributed solely to the tokenization strategy.

The independent variable in this study is the tokenization method (standard BPE vs. hybrid BPE), while the dependent variables are divided into two categories. Intrinsic evaluation metrics—such as word fragmentation rate, token fertility, and morph-boundary violation rate—are used to assess the structural quality of the generated subword units.

Metric	Definition	Malayalam Context
Word Fragmentation Rate (WFR)	% of words split into more than one token	High for Malayalam due to agglutinative morphology
Average Token Fertility	Tokens produced per word	Indicates sequence length inflation
Characters Per Token (CPT)	Average characters represented per token	Higher values indicate larger linguistic units
Single Token Retention Rate (STRR)	% of frequent words preserved as single tokens	Low values suggest vocabulary bias against long Malayalam words
Tokenization Parity (TP)	Ratio of tokenized length across languages	Malayalam typically requires more tokens than English
Information Parity (IP)	Relative compression efficiency across languages	Strong predictor of downstream semantic task performance
Morph-Boundary Violation Rate	Alignment between token and morpheme boundaries	Measures linguistic fidelity using Precision, Recall, and F1

Table 5: Evaluation metrics for analyzing tokenization behavior in Malayalam

Extrinsic evaluation is performed using Accuracy and Macro-F1 score on a topic classification task, reflecting the effectiveness of the representation in a real-world NLP scenario.

The experimental workflow is structured in three stages. First, a linguistic rule framework is defined to capture high-frequency morphological patterns in Malayalam, including suffix stacking and verbal inflections. Second, intrinsic evaluation is conducted to measure how each tokenizer handles these structures. Finally, both tokenization schemes are evaluated within a neural classification pipeline to determine their impact on semantic understanding.

To assess the robustness of the findings, each experiment is repeated across multiple runs with controlled random seeds, and results are reported as mean and standard deviation. In addition, statistical significance testing is applied to determine whether observed differences between the two tokenization approaches are meaningful rather than due to random variation.

Overall, this research design isolates the effect of morphology-aware tokenization and provides a systematic framework for evaluating its contribution to both representation efficiency and downstream model performance in Malayalam NLP.

Framework Task	Implementation Goal	Evaluation Metric
Rule Formalization	Identify root and suffix boundaries	Boundary Precision/Recall
Hybrid Tokenizer Design	Constrain BPE merge steps	Morphological Alignment (MorphScore)
Intrinsic Evaluation	Quantify fragmentation changes	Word Fragmentation Rate (WFR), Token Fertility
Downstream Task	Train topic classifier	Macro-F1, convergence stability

Table 6: Framework tasks, implementation goals, and evaluation metrics

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